

# Project Statement

## Personal Finance Manager

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### Problem Statement

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Many students and young people do not keep track of their money. Income such as pocket money or a stipend and expenses such as food, books and travel are usually remembered only roughly, so it is hard to know how much money is actually left. Notebooks and spreadsheets work, but they take effort to maintain, and paper records are hard to search.

There is a need for a simple, lightweight tool that lets a person record income and expenses quickly, see the current balance at any time, and find past transactions without any setup or installation beyond Python.

### Objective

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To build a command-line Personal Finance Manager in Python that lets one or more users record their income and expenses, view their transaction history, check their balance, and search past transactions by keyword.

### Scope of the Project

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#### Included

- Creating multiple users, each with a separate list of transactions
- Logging in to an existing user
- Adding income and expense transactions with an amount and a description
- Viewing all transactions of a user
- Calculating and displaying the current balance (total income minus total expenses)
- Searching transactions by keyword in the description (case-insensitive)
- Input validation for menu choices, transaction type, amounts and text fields

#### Not included (out of scope for this version)

- Permanent storage (data is kept in memory and is lost when the program exits)
- Passwords or any form of authentication beyond username lookup
- Editing or deleting transactions
- Categories, budgets, charts or reports
- A graphical or web interface

### Target Users

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- Students who want a quick way to track pocket money and daily expenses
- Beginners who want a simple example of a menu-driven Python program
- Anyone who prefers a small offline tool over a full finance application

## High-Level Features

| No. | Feature           | Description                                                           |
|-----|-------------------|-----------------------------------------------------------------------|
| 1   | User management   | Create users and log in with a username                               |
| 2   | Add transaction   | Record an income or expense with amount and description               |
| 3   | View transactions | Display a numbered list of all transactions                           |
| 4   | Check balance     | Show the current balance in ₹                                         |
| 5   | Search            | Find transactions containing a keyword                                |
| 6   | Input validation  | Re-prompt on empty, invalid or non-positive input instead of crashing |

## Expected Outcome

A working, menu-driven Python program that runs in any terminal with no external libraries. A user can create an account, add several income and expense entries, see an accurate balance, and search their history, with the program handling incorrect input gracefully.

## Technologies Used

| Item         | Details                                                                        |
|--------------|--------------------------------------------------------------------------------|
| Language     | Python 3                                                                       |
| Interface    | Command line (text menus)                                                      |
| Data storage | In-memory dictionary of users, each holding a list of transaction dictionaries |
| Libraries    | None (built-in Python only)                                                    |

## Limitations

- All data is lost when the program is closed.
- Usernames are not protected by passwords.
- Transactions cannot be modified or removed once added.

## Future Enhancements

- Save and load data using JSON, CSV or a database
- Password-protected user accounts
- Edit and delete transactions
- Spending categories and monthly summaries
- Unit tests and a graphical interface

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